

# **SEMESTER 7**

**INFORMATION TECHNOLOGY**

## SEMESTER S7

### NATURAL LANGUAGE PROCESSING

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT741</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

- Students will be able to understand fundamental concepts of natural language processing (NLP).
- Students will be able to understand advanced NLP techniques for syntactic and semantic understanding, including POS tagging, HMM tagging, named entity recognition, coreference resolution, and semantic role labelling
- Students will be able to understand practical NLP applications such as machine translation, question answering systems, chatbots, dialogue systems, automatic speech recognition, and text-to-speech technologies.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Contact Hours</b> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Introduction to Natural Language Understanding: Applications, Levels of NLP, Basic Text Processing, Empirical Laws, Minimum Edit Distance.<br>N-gram Language Models: N-grams, Evaluating Language Models, Smoothing:Knesser-Ney Smoothing, Perplexity's Relation to Entropy.<br>Naive Bayes, Text Classification, and Sentiment: Naive Bayes Classifiers, Training the Naive Bayes Classifier, Evaluation: Precision, Recall, F-measure, Test sets and Cross-validation. | <b>9</b>             |
| <b>2</b>          | POS Tagging, HMM POS Tagging, Viterbi Algorithm, Morphology, Named Entity Recognition, Conditional Random Fields.                                                                                                                                                                                                                                                                                                                                                         | <b>9</b>             |

|          |                                                                                                                                                                                                      |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | Lexical Semantics, Vector Semantics, TF-IDF, PMI, Word2Vec, Word Senses, Word Sense Disambiguation, WordNet<br>Context-Free Grammars and Constituency Parsing, Treebank.                             |          |
| <b>3</b> | Semantic Roles, Logical Representations of Sentence Meaning, Semantic Role Labeling, Lexicons for Sentiment, Affect, and Connotation, Coreference Resolution, Discourse Coherence.                   | <b>9</b> |
| <b>4</b> | NLP Applications:<br>Machine Translation, Question Answering and Information Retrieval, Chatbots & Dialogue Systems, Automatic Speech Recognition and Text-to-Speech, Relation and Event Extraction. | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Micro project</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|--------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                            | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| <b>Part A</b>                                                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                                                                      | <b>Total</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                             | Bloom's Knowledge Level (KL) |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Explain and implement basic text processing, n-gram language models, and Naive Bayes classifiers                                                                            | <b>K2</b>                    |
| <b>CO2</b>     | Illustrate advanced natural language processing techniques to analyze and process natural language text effectively.                                                        | <b>K2</b>                    |
| <b>CO3</b>     | Describe semantic role labeling, logical representations of sentence meaning, and coreference resolution to ensure discourse coherence and accurate semantic understanding. | <b>K2</b>                    |
| <b>CO4</b>     | Summarize different natural language processing (NLP) applications by applying appropriate NLP techniques and evaluating their effectiveness.                               | <b>K2</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | 2   | -   | 2   | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO2</b> | 3   | 2   | 2   | -   | 2   | -   | -   | -   | -   | -    | -    | -    |
| <b>CO3</b> | 3   | 2   | 2   | -   | 2   | -   | -   | -   | -   | -    | -    | -    |
| <b>CO4</b> | 3   | 2   | 2   | -   | 2   | 2   |     | 2   | -   | -    | -    | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                |                                 |                              |                         |
|-------------------|--------------------------------|---------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>       | <b>Name of the Author/s</b>     | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Speech and Language Processing | Dan Jurafsky,<br>James H Martin | Pearson Education<br>India   | 3 <sup>rd</sup> edition |

| <b>Reference Books</b> |                                                        |                             |                              |                         |
|------------------------|--------------------------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                               | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Natural Language Understanding                         | James Allen                 | Pearson                      | 2 <sup>nd</sup> edition |
| 2                      | Foundations of Statistical Natural Language Processing | Christopher Manning         | MIT Press                    | 1 <sup>st</sup> edition |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                                              |
| 1                                     | <a href="https://onlinecourses.nptel.ac.in/noc23_cs45/preview">https://onlinecourses.nptel.ac.in/noc23_cs45/preview</a>                     |
| 2                                     | <a href="https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs56/">https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs56/</a> |
| 3                                     | <a href="https://web.stanford.edu/~jurafsky/slp3/">https://web.stanford.edu/~jurafsky/slp3/</a>                                             |

## SEMESTER S7

### SOFTWARE DEVELOPMENT WITH AGILE AND DEVOPS

|                                            |                                |                    |                |
|--------------------------------------------|--------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT742</b>                | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                        | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                              | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PCITT503: Software Engineering | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To understand the core concepts of DevOps and how they integrate with Agile practices to enhance software development and deployment.
2. To understand the core concepts of DevOps and how they integrate with Agile practices to enhance software development and deployment.
3. To equip students with the knowledge and skills to develop secure software by integrating security best practices throughout the development lifecycle.
4. To explore advanced concepts in Agile and DevOps, with a focus on scaling and continuous improvement in large and complex environments.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Introduction to Agile Development</b><br>Introduction to Agile Methodology: Overview of traditional vs. Agile development, Agile Manifesto and Principles.<br>Agile Frameworks: Scrum, Kanban, XP (Extreme Programming).<br>Agile Project Management: User Stories, Backlogs, Sprints, and Releases.<br>Roles in Agile Teams: Product Owner, Scrum Master, Development Team.<br>Agile Planning and Estimation: Story Points, Velocity, and Burndown Charts.<br>Agile Metrics and Reporting: Measuring progress, continuous improvement. | <b>9</b>             |
| <b>2</b>          | <b>DevOps Fundamentals and Practices</b><br>Introduction to DevOps: Definition, need for DevOps, DevOps lifecycle.                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | <p>Continuous Integration and Continuous Deployment (CI/CD): Jenkins, GitLab CI, Travis CI.</p> <p>Infrastructure as Code (IaC): Terraform, Ansible, Docker, and Kubernetes.</p> <p>Version Control Systems: Git, Git workflows, branching strategies.</p> <p>Automated Testing in DevOps: Unit tests, integration tests, deployment testing.</p>                                                                                                                                                                                                               |          |
| <b>3</b> | <p><b>Secure Software Development</b></p> <p>Introduction to Secure Software Development: Importance of security in software, common vulnerabilities.</p> <p>Secure Coding Practices: OWASP Top 10, secure coding standards.</p> <p>Threat Modeling: Identifying threats, security design principles.</p> <p>Security Testing: Static Application Security Testing (SAST), Dynamic Application Security Testing (DAST), and penetration testing.</p> <p>Security in CI/CD Pipeline: Integrating security into DevOps (DevSecOps), automated security tools.</p> | <b>9</b> |
| <b>4</b> | <p><b>Advanced DevOps and Agile Practices</b></p> <p>Advanced CI/CD Pipelines: Custom pipelines, Blue-Green deployments, Canary releases.</p> <p>Monitoring and Logging: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana).</p> <p>Cloud and DevOps: DevOps practices in AWS, Azure, and GCP.</p> <p>Agile at Scale: Scaling Agile in large organizations, SAFe (Scaled Agile Framework), LeSS (Large Scale Scrum).</p> <p>Continuous Feedback and Improvement: Retrospectives, continuous learning and adaptation.</p>                          | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

### End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| Part A                                                                                                                                                      | Part B                                                                                                                                                                                                                                                                      | Total     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p>(8x3 =24marks)</p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 sub divisions.</li></ul> <p>(4x9 = 36 marks)</p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                              | Bloom's Knowledge Level (KL) |
|----------------|--------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Discuss Agile principles and practices to manage and deliver software projects effectively.                  | <b>K2</b>                    |
| <b>CO2</b>     | Explain DevOps practices to automate and streamline software development, testing, and deployment processes. | <b>K2</b>                    |
| <b>CO3</b>     | Describe secure coding practices and integrate security into the software development process.               | <b>K2</b>                    |
| <b>CO4</b>     | Illustrate advanced DevOps practices in a cloud environment to enhance software delivery.                    | <b>K2</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | 3   | -   | -   | -   | -   | -   | -   | -    | 2    | 2    |
| <b>CO2</b> | 3   | 3   | 3   | -   | -   | -   | -   | -   | -   | -    | 2    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 2   | -   | -   | -   | -   | -   | -    | 2    | 2    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 2   | -   | -   | -   | -   | -    | 2    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation



| Text Books |                                                                 |                      |                       |                   |
|------------|-----------------------------------------------------------------|----------------------|-----------------------|-------------------|
| Sl. No     | Title of the Book                                               | Name of the Author/s | Name of the Publisher | Edition and Year  |
| 1          | Agile Software Development, Principles, Patterns, and Practices | Robert C. Martin     | Prentice Hall         | 1st Edition, 2002 |

| Reference Books |                                                                                                |                                       |                       |                               |
|-----------------|------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------|-------------------------------|
| Sl. No          | Title of the Book                                                                              | Name of the Author/s                  | Name of the Publisher | Edition and Year              |
| 1               | Scrum: The Art of Doing Twice the Work in Half the Time                                        | Jeff Sutherland                       | Crown Business        | 1st Edition, 2014             |
| 2               | The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win                   | Gene Kim, Kevin Behr, George Spafford | IT Revolution Press   | 5th Anniversary Edition, 2018 |
| 3               | The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws                   | Dafydd Stuttard, Marcus Pinto         | Wiley                 | 2nd Edition, 2011             |
| 4               | Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation | Jez Humble, David Farley              | Addison-Wesley        | 1st Edition, 2010             |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                                                                                                       |
| 1                              | <a href="https://www.youtube.com/watch?v=VFQtSqChlsk&amp;t=4223s">https://www.youtube.com/watch?v=VFQtSqChlsk&amp;t=4223s</a>                                                                 |
| 2                              | <a href="https://www.youtube.com/watch?v=hQcFE0RD0cQ">https://www.youtube.com/watch?v=hQcFE0RD0cQ</a>                                                                                         |
| 3                              | <a href="https://www.youtube.com/watch?v=ONe_P09MSKM">https://www.youtube.com/watch?v=ONe_P09MSKM</a>                                                                                         |
| 4                              | <a href="https://www.youtube.com/watch?v=egTi9U9vw3E&amp;list=PLjNII-JkdjFz5EXWIGMBRk63PC8uJsHMo">https://www.youtube.com/watch?v=egTi9U9vw3E&amp;list=PLjNII-JkdjFz5EXWIGMBRk63PC8uJsHMo</a> |

## SEMESTER S7

### BLOCKCHAIN TECHNOLOGY

|                                            |                                             |                    |                |
|--------------------------------------------|---------------------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT743</b>                             | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                                     | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                                           | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PCITT601: Cryptography and Network Security | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. Students will be familiar with the working of popular blockchain platforms: bitcoin & Ethereum and will acquire understanding of underlying principles of blockchain.
2. Students will be able to develop and deploy smart contracts in Ethereum blockchain.

#### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Contact Hours</b> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Blockchain Introduction:</b> <i>Fundamentals of blockchain</i> - historical background, basic terminologies in Blockchain, structure of a block ( <i>till this from Ref. 3</i> ), Blockchain architecture, generic elements of a blockchain, How blockchain works, benefits, features, and limitations of blockchain ( <i>till this from Textbook 1</i> ), Types of Blockchain, The Evolutionary Transformation of Blockchain, Comparison of different generation of blockchain ( <i>till this from Ref. 3</i> )<br><b>Decentralization using blockchain:</b> Methods of decentralization, Routes to decentralization, Blockchain and full ecosystem decentralization, IPFS basics | <b>8</b>             |
| <b>2</b>          | <b>Cryptography for blockchain:</b> cryptographic primitives - Secure Hash algorithms, Applications of cryptographic hash functions, Message Authentication Codes, Digital signatures, Merkle tree, Elliptic curve cryptography fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          | <b>Consensus Algorithms:</b> Introducing the consensus problem, The Byzantine generals problem, Fault tolerance, FLP impossibility, Classification and categories of consensus algorithms - CFT algorithms, BFT algorithms, Nakamoto consensus, Proof of Stake, Choosing a consensus algorithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
| <b>3</b> | <b>Bitcoin:</b> The beginnings of Bitcoin, Bitcoin Definition, Bitcoin - a user's perspective, Cryptographic keys, Addresses in Bitcoin, Transactions, The transaction lifecycle, The transaction data structure, The Script language, Types of scripts, Coinbase transactions, Transaction validation, Transaction bugs, Bitcoin blockchain, The genesis block, Mining, Proof of work, The mining algorithm, The hash rate, Mining systems, Mining pools, wallets                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>7</b>  |
| <b>4</b> | <p><b>Ethereum:</b> A General-Purpose Blockchain, Ethereum's components, Ethereum and Turing Completeness, From General-Purpose Blockchains to Decentralized Applications (DApps), Ether currency units, Getting Started with MetaMask, Introducing the world Computer, Externally Owned Accounts (EOAs) and Contracts</p> <p><b>A simple Contract:</b> A Test Ether Faucet, Compiling the Faucet Contract, Creating the Contract on the Blockchain, Interacting with the Contract, The Structure of a Transaction, Special Transaction: Contract Creation, A Blockchain Based Supply Chain Management Testcase ( from Ref. 3)</p> <p><b>Smart Contracts and Solidity:</b> Life Cycle of a Smart Contract, Building a Smart Contract with Solidity, The Ethereum Contract ABI, Programming with Solidity</p> <p><b>Tokens:</b> How Tokens Are Used, The ERC20 Token Standard, Launching Our Own ERC20 Token,</p> | <b>12</b> |

#### Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

#### Continuous Internal Evaluation Marks (CIE):

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written) | Total |
|------------|-----------------------------|----------------------------------------|-----------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                      | 40    |

### End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| Part A                                                                                                                                                      | Part B                                                                                                                                                                                                                                                                      | Total     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p>(8x3 =24marks)</p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 sub divisions.</li></ul> <p>(4x9 = 36 marks)</p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                      | Bloom's Knowledge Level (KL) |
|----------------|----------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | summarize foundational concepts of decentralization with blockchain  | <b>K2</b>                    |
| <b>CO2</b>     | Interpret the importance of cryptography and consensus in Blockchain | <b>K2</b>                    |
| <b>CO3</b>     | Outline the working of Bitcoin blockchain                            | <b>K2</b>                    |
| <b>CO4</b>     | Build and deploy smart contract in Ethereum blockchain               | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | -   | 3   | -   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO2</b> | 3   | -   | -   | 3   | 2   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO3</b> | 3   | 3   | 3   | 2   | 2   | -   | -   | -   | -   | -    | -    | 3    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 3   | -   | -   | -   | -   | -    | -    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                      |                                               |                       |                                 |
|------------|----------------------|-----------------------------------------------|-----------------------|---------------------------------|
| Sl. No     | Title of the Book    | Name of the Author/s                          | Name of the Publisher | Edition and Year                |
| 1          | Mastering Blockchain | Imran Bashir                                  | Packt Publishing Ltd. | 3 <sup>rd</sup> Edition<br>2020 |
| 2          | Mastering Ethereum   | Andreas M. Antonopoulos<br>and Dr. Gavin Wood | O'Reilly Media, Inc   | 1 <sup>st</sup> Edition<br>2018 |

| Reference Books |                                                                                           |                                             |                       |                                 |
|-----------------|-------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------|---------------------------------|
| Sl. No          | Title of the Book                                                                         | Name of the Author/s                        | Name of the Publisher | Edition and Year                |
| 1               | Mastering Blockchain                                                                      | Lorne Lantz and Daniel Cawrey               | O'Reilly Media, Inc., | 1 <sup>st</sup> Edition<br>2020 |
| 2               | Solidity Programming Essentials                                                           | Ritesh Modi                                 | Packt Publishing      | 1 <sup>st</sup> Edition<br>2018 |
| 3               | Blockchain 1.0 to Blockchain 4.0—The Evolutionary Transformation of Blockchain Technology | Pratyusa Mukherjee,<br>Chittaranjan Pradhan | De Gruyter            | May, 2021                       |
| 4               | Bitcoin: A Peer-to-Peer Electronic Cash System                                            | Satoshi Nakamoto                            | Bitcoin.org           | Oct, 2008                       |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                                     |
| 1                              | <a href="https://nptel.ac.in/courses/106105235">https://nptel.ac.in/courses/106105235</a>                                   |
| 2                              | <a href="https://nptel.ac.in/courses/106105235">https://nptel.ac.in/courses/106105235</a>                                   |
| 3                              | <a href="https://nptel.ac.in/courses/106105235">https://nptel.ac.in/courses/106105235</a>                                   |
| 4                              | <a href="https://onlinecourses.swayam2.ac.in/aic21_ge01/preview">https://onlinecourses.swayam2.ac.in/aic21_ge01/preview</a> |

## SEMESTER S7

### MOBILE APP DEVELOPMENT

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT744</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 2:0:1:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce students to the fundamentals of mobile application development, including different mobile platforms and development environments.
2. To familiarize students with UI/UX design principles and their application in creating user-friendly and aesthetically pleasing mobile applications.
3. To provide knowledge on backend integration, including APIs, cloud services, and database management for mobile applications.
4. To explore advanced topics in mobile app development such as performance optimization, security practices, cross-platform development, and future trends.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Contact Hours</b> |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <p>Introduction to Mobile App Development - Overview of Mobile Application Development: History, Trends, and Importance, Mobile Platforms: Android, iOS, and Cross-Platform Development, Development Environments: Android Studio, Xcode, and Visual Studio Code, Basic App Development Lifecycle: Design, Development, Testing, and Deployment, Introduction to Programming Languages: Java, Kotlin, Swift, and Dart.</p> <p><b>Lab:</b> Setting up development environments (Android Studio, Xcode, Visual Studio Code), Building a simple "Hello World" app using Java (Android) and Swift (iOS), Exploring cross-platform development with Flutter</p> | <b>11</b>            |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2 | <p>UI/UX Design for Mobile Apps - UI/UX Design Principles: Visual Design, Information Architecture, Interaction Design - Wireframing and Prototyping Tools: Figma, Sketch, Adobe XD - Responsive Design for Multiple Devices and Screen Sizes - Material Design Guidelines for Android and Human Interface Guidelines for iOS - Accessibility in Mobile App Design</p> <p><b>Lab:</b> Creating wireframes and prototypes using Figma or Sketch - Implementing responsive design in Android and iOS apps - Applying Material Design and Human Interface Guidelines in simple UI components -Testing app accessibility features on various devices</p>      | 11 |
| 3 | <p>Backend Integration and APIs- Connecting Mobile Apps to Backend Services: RESTful APIs, GraphQL-Cloud Services for Mobile Apps: Firebase, AWS Amplify, Google Cloud-Working with Databases: SQLite, Realm, Firebase Realtime Database-Data Storage and Management in Mobile Apps-Implementing Push Notifications and In-App Messaging</p> <p><b>Lab:</b> Setting up and integrating a Firebase backend with a mobile app-Implementing RESTful API calls in an app using Retrofit (Android) and URLSession (iOS)-Storing and retrieving data using SQLite and Firebase Realtime Database-Configuring push notifications and sending in-app messages</p> | 11 |
| 4 | <p>Advanced Topics in Mobile App Development-Performance Optimization: Profiling, Memory Management, and Battery Efficiency-Mobile App Security: Secure Coding Practices, Encryption, and Authentication-Cross-Platform Development: React Native, Flutter, Xamarin-App Store Deployment: Google Play Store, Apple App Store.</p> <p><b>Lab:</b> Profiling and optimizing app performance using Android Studio and Xcode tools, Implementing secure coding practices and encryption in mobile apps, Developing and deploying a cross-platform app using Flutter, Preparing and submitting apps to the Google Play Store and Apple App Store.</p>          | 11 |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| <b>Part A</b>                                                                                                                                                                                  | <b>Part B</b>                                                                                                                                                                                                                                                                                                  | <b>Total</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 sub divisions.</li></ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |



### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                    | Bloom's Knowledge Level (KL) |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Demonstrate an understanding of mobile application development fundamentals, including platform architecture and development environments.         | <b>K3</b>                    |
| <b>CO2</b>     | Apply UI/UX design principles to develop user-centric mobile applications with effective interface designs.                                        | <b>K3</b>                    |
| <b>CO3</b>     | Use backend services, APIs, and databases in mobile applications to enhance functionality and data management.                                     | <b>K3</b>                    |
| <b>CO4</b>     | Implement mobile applications by optimizing for performance and security, and apply cross-platform development strategies and future technologies. | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 2   | -   | 2   | -   | -   | -   | 3   | -    | -    | -    |
| <b>CO2</b> | 2   | 3   | 2   | 2   | 2   | -   | -   | -   | 3   | -    | -    | -    |
| <b>CO3</b> | 2   | 3   | 3   | 2   | 2   | -   | -   | -   | 3   | 2    | -    | -    |
| <b>CO4</b> | 3   | 3   | 3   | 2   | 3   | -   | -   | -   | 3   | 2    | -    | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                               |                                                 |                              |                                 |
|-------------------|-----------------------------------------------|-------------------------------------------------|------------------------------|---------------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                      | <b>Name of the Author/s</b>                     | <b>Name of the Publisher</b> | <b>Edition and Year</b>         |
| 1                 | Android Programming: The Big Nerd Ranch Guide | Bill Phillips, Chris Stewart, Kristin Marsicano | Big Nerd Ranch Guides        | 4 <sup>th</sup> Edition<br>2019 |
| 2                 | iOS Programming: The Big Nerd Ranch Guide     | Christian Keur, Aaron Hillegass                 | Big Nerd Ranch Guides        | 7 <sup>th</sup> Edition<br>2020 |

| <b>Reference Books</b> |                                                                                                          |                              |                              |                                 |
|------------------------|----------------------------------------------------------------------------------------------------------|------------------------------|------------------------------|---------------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                                                                 | <b>Name of the Author/s</b>  | <b>Name of the Publisher</b> | <b>Edition and Year</b>         |
| 1                      | Flutter for Beginners: An Introductory Guide to Building Cross-Platform Mobile Applications              | Alessandro Biessek           | Packt Publishing             | 2 <sup>nd</sup> Edition<br>2021 |
| 2                      | Kotlin Programming: The Big Nerd Ranch Guide                                                             | Josh Skeen, David Greenhalgh | Big Nerd Ranch Guides        | 1 <sup>st</sup> Edition<br>2018 |
| 3                      | Pro iOS 14 Development with Swift 5                                                                      | Wallace Wang                 | Apress                       | 2021<br>Bonnie Eisenman         |
| 4                      | Learning React Native: Building Native Mobile Apps with JavaScript                                       | Bonnie Eisenman              | O'Reilly Media               | 2 <sup>nd</sup> Edition<br>2017 |
| 5                      | Mobile App Development with Ionic, Revised Edition: Cross-Platform Apps with Ionic, Angular, and Cordova | Chris Griffith               | O'Reilly Media               | 2017                            |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                              |
| 1                                     | <a href="https://onlinecourses.swayam2.ac.in/nou21_ge41/preview">https://onlinecourses.swayam2.ac.in/nou21_ge41/preview</a> |
| 2                                     | <a href="https://archive.nptel.ac.in/courses/106/106/106106156/">https://archive.nptel.ac.in/courses/106/106/106106156/</a> |

## SEMESTER S7

### DEEP LEARNING

|                                            |                            |             |                |
|--------------------------------------------|----------------------------|-------------|----------------|
| <b>Course Code</b>                         | <b>PEITT745</b>            | CIE Marks   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                    | ESE Marks   | 60             |
| <b>Credits</b>                             | 5/3                        | Exam Hours  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PCITT501: Machine Learning | Course Type | Theory         |

#### Course Objectives:

1. To enable students to understand the fundamental deep learning techniques and apply them to solve real-world problems.
2. To enable students to analyze and evaluate advanced deep learning models and their applications to determine the most suitable approach for complex problems.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                       | <b>Contact Hours</b> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Deep Learning Fundamentals:</b> Principal Component Analysis, Conditional Probability- Chain Rule, Bayes' Rule, Hyper parameter Optimization - Gradient- Based Optimization – SGD, Adaptive GD, RMSprop, Constrained Optimization, Regularization - L1 and L2 regularization.<br><b>Deep Feedforward Networks:</b> Gradient-Based Learning, Hidden Units, Architecture Design, Backpropagation | <b>9</b>             |
| <b>2</b>          | <b>Recurrent Neural Networks:</b><br>Architecture, Challenges, Bi-directional RNN, Encoder- Decoder Sequence-to-Sequence Application, LSTM, Gated Recurrent Unit networks, Applications of RNN.                                                                                                                                                                                                   | <b>9</b>             |

|          |                                                                                                                                                                                                                                                  |          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>3</b> | <b>Convolutional Networks:</b><br>Basic Structure of Convolutional Network, Training a Convolutional Network, Transfer learning models - AlexNet, GoogleNet, VGG Net, ResNet, Applications of CNN                                                | <b>9</b> |
| <b>4</b> | <b>Advanced Deep Learning:</b><br>Transformers and Pre-trained networks, Generative Adversarial Network, Autoencoder: Basic Principles, Deep Autoencoder, Applications, Competitive Learning- Self Organising Map, Applications of Deep Learning | <b>9</b> |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Internal Examination (Written)</b> | <b>Evaluate Level Assessment</b> | <b>Analyze Level Assessment</b> | <b>Total</b> |
|-------------------|---------------------------------------|----------------------------------|---------------------------------|--------------|
| <b>5</b>          | <b>15</b>                             | <b>10</b>                        | <b>10</b>                       | <b>40</b>    |

**Evaluate and Analyze Level Assessment [20 Marks]**

Students should evaluate and analyze a real-world problem, assess the proposed solutions, provide a conclusion on which solution is most appropriate for the problem, and implement the chosen solution.

**Criteria for evaluation:**

**1. Problem Definition (K2 - 4 points)**

- a. Clearly defines the real-world problem, ensuring it is specific and relevant.*
- b. Problem is excellently defined with precise, impactful objectives that highlight the importance and scope of the issue.*

**2. Problem Analysis (K4 - 4 points)**

- a. Break-down and presents a well-reasoned solution approach, including the rationale behind chosen methods.*
- b. Performance metrics are expertly selected, analysis is deep, insightful, and*

*demonstrates strong understanding of the problem and proposed solutions.*

**3. Evaluate (K5 - 4 points)**

- a. Thoroughly evaluate the proposed solutions, comparing their effectiveness and relevance.*
- b. Data is excellently selected; preprocessing is sophisticated and enhances model performance.*
- c. Training process is expertly handled, validation is extensive, demonstrating strong generalization and robustness of the model.*

**4. Implementation (K3 - 4 points)**

- a. Select the most feasible solution based on evaluation and implements it.*
- b. Successfully translates the chosen solution into code with a clear, working implementation.*
- c. Design is innovative, implementation is flawless and optimized for performance.*

**5. Conclusion (K4- 2 points, K5 – 2 points)**

- a. Provides a highly creative and innovative conclusion, pushing the boundaries of standard methods. (K4)*
- b. Demonstrates an excellent grasp of deep learning concepts, applied creatively and effectively in the context of the solution. (K5)*

**Scoring:**

- 1. Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
- 2. Competent (3 points):** Solid performance with minor areas for improvement.
- 3. Developing (2 points):** Adequate effort but lacks depth or clarity in analysis or implementation.
- 4. Minimal (1 point):** Incomplete or significantly flawed analysis or implementation.

### End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| Part A                                                                                                                                                      | Part B                                                                                                                                                                                                                                                                      | Total     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p>(8x3 =24marks)</p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 sub divisions.</li></ul> <p>(4x9 = 36 marks)</p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                    | Bloom's Knowledge Level (KL) |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Apply fundamental concepts of deep learning to design, train, and evaluate deep feedforward networks.                                              | <b>K3</b>                    |
| <b>CO2</b>     | Implement and compare various gradient-based optimization techniques to optimize deep learning models.                                             | <b>K3</b>                    |
| <b>CO3</b>     | Design and develop deep feedforward networks and recurrent neural networks, including architectures like LSTM and GRUs, to address specific tasks. | <b>K3</b>                    |
| <b>CO4</b>     | Explore and apply various CNN to solve complex visual recognition tasks.                                                                           | <b>K3</b>                    |
| <b>CO5</b>     | Analyze and evaluate advanced deep learning techniques, to innovative applications.                                                                | <b>K5</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   | 2   | 2   | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO2</b> | 3   | 2   | 3   | 3   | 2   | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 3   | 2   | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 3   | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO5</b> | 3   | 3   | 3   | 3   | 3   | 3   | -   | -   | 3   | 3    | 2    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                               |                                    |                                                                                                 |                         |
|-------------------|-----------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                      | <b>Name of the Author/s</b>        | <b>Name of the Publisher</b>                                                                    | <b>Edition and Year</b> |
| 1                 | Neural Networks and Deep Learning: A Textbook | Charu C. Aggarwal                  | Springer International Publishing AG                                                            | 1st ed. 2018            |
| 2                 | Speech and Language Processing                | James H Matin, Dan Jurafsky        | <a href="https://web.stanford.edu/~jurafsky/slp3/">https://web.stanford.edu/~jurafsky/slp3/</a> | (3rd ed. draft)         |
| 3                 | Deep Learning: Foundations and Concepts       | Christopher M. Bishop, Hugh Bishop | Springer-Nature New York Inc                                                                    | 2 November 2023         |

| <b>Reference Books</b> |                                                                  |                                                |                              |                                           |
|------------------------|------------------------------------------------------------------|------------------------------------------------|------------------------------|-------------------------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                         | <b>Name of the Author/s</b>                    | <b>Name of the Publisher</b> | <b>Edition and Year</b>                   |
| 1                      | Machine Learning: Theory and Practice                            | M. Narasimha Murty, Ananthanarayana V S        | Univerities Press            | 1 <sup>st</sup> Edition, 2024             |
| 2                      | Deep Learning (Adaptive Computation and Machine Learning Series) | Ian Goodfellow, Yoshua Bengio, Aaron Courville | MIT Press                    | 2016                                      |
| 3                      | Pattern Recognition and Machine Learning                         | Bishop C M                                     | Springer                     | 1 <sup>st</sup> Edition Reprint, Jan 2009 |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                                 |
| 1                              | <a href="https://onlinecourses.nptel.ac.in/noc21_cs76/preview">https://onlinecourses.nptel.ac.in/noc21_cs76/preview</a> |
| 2                              | <a href="https://onlinecourses.nptel.ac.in/noc20_cs62/preview">https://onlinecourses.nptel.ac.in/noc20_cs62/preview</a> |



## SEMESTER S7

### APPROXIMATION ALGORITHMS

|                                            |                                         |                    |                |
|--------------------------------------------|-----------------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT751</b>                         | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                                 | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                                       | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PCITT502: Algorithm Analysis and Design | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce the fundamental concepts of approximation algorithms and their importance in solving NP-hard problems.
2. To explore various design techniques used in approximation algorithms, including greedy methods, layering, primal-dual method and semidefinite programming.
3. To understand the analysis of approximation algorithms and the concepts of approximation ratio, performance guarantees and inapproximability results.
4. To introduce the fundamental concepts of fixed-parameter tractable algorithms and various design techniques used in parameterized complexity to solve NP-hard problems.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Module 1: Overview of Computational Complexity and Introduction to Approximation Algorithms</b><br><b>Overview of Computational Complexity:</b> Decision problem vs Optimization problem - P, NP, co-NP, NP-Hard, and NP-Complete classes—along with their interrelationships- Intractability of NP-Complete problems: Challenges and Importance- Polynomial time reductions- Examples of NP-Complete problems -SAT, 3-SAT, Clique, Vertex Cover.<br><b>Approaches to solve NP-Complete Problems (Concept only):</b> Approximation algorithms, Exact algorithms, Randomized algorithms, | <b>9</b>             |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <p>Fixed-parameter tractable algorithms.</p> <p><b>Introduction to Approximation Algorithms:</b> Definition and Motivation- Approximation Ratio: Definition - Basic Techniques : <i>Greedy Algorithms and Local Search</i>. : Basic Concept, Algorithm and analysis of approximation factor (detailed analysis not required) of Set Cover problem, Scheduling jobs on identical parallel machines, Travelling Salesman problem, Metric Steiner Tree problem.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| 2 | <p><b>Module 2: Advanced Techniques for Designing Approximation Algorithms</b></p> <p><b>Primal Dual Schema:</b> Introduction to Linear Programming, Concept of Primal Dual Schema- LP Duality Theorem, Complementary Slackness Theorem. Linear programming formulation of the Set Cover, Vertex Cover, Minimum Multicut, Multiwaycut and Weighted Vertex Cover Problems - Approximation algorithm for the SET Cover problem - using simple rounding technique and primal dual schema - analysis of approximation factor.</p> <p><b>Randomized Algorithms:</b> Concept of randomization, Randomized rounding -Set Cover Problem, Randomized algorithm for MAX CUT problem, Derandomization of MAX CUT algorithm.</p> <p><b>Semidefinite Programming (SDP) :</b> Basics of SDP and its relation to LP, Formulating approximation problems using SDP, SDP-based approximation algorithm and analysis of approximation ratio for the MAX-CUT problem.</p> | 9 |
| 3 | <p><b>Module 3: Hardness of Approximation Algorithms</b></p> <p><b>Hardness of Approximation:</b> Reductions, gaps, and hardness factors, PCP Theorem, Hardness of MAX-3SAT, Hardness of MAX-3SAT with bounded occurrence of variables, Hardness of vertex cover and Steiner tree, Hardness of Clique.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 9 |
| 4 | <p><b>Module 4: Introduction to Parameterized Complexity Theory</b></p> <p><b>Introduction to Parameterized Complexity Theory :</b> Parameterized problem- Definition and Example, Fixed-parameter Tractable (FPT) algorithms - Definition.</p> <p><b>Design Techniques in Parameterized Complexity :</b> <i>Kernelization</i>- Concept, FPT algorithms for Vertex Cover problem and 3-Hitting set problem, <i>Bounded Search Tree</i> - Concept, FPT algorithms for Vertex Cover problem and Feedback Vertex Set problem, <i>Iterative Compression</i> - Concept, FPT algorithms for Vertex Cover problem.</p> <p><b>Parameterized Intractability:</b> Parameterized reductions, W-hierarchy.</p>                                                                                                                                                                                                                                                     | 9 |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written) | Total |
|------------|-----------------------------|----------------------------------------|-----------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                      | 40    |

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| Part A                                                                                                                                                         | Part B                                                                                                                                                                                                                                                                          | Total |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p>(8x3 =24marks)</p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p>(4x9 = 36 marks)</p> | 60    |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                         | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>CO1</b>     | Understand the fundamentals of computational complexity and apply the basic concepts of approximation algorithms to solve NP-hard problems with near-optimal solutions. | <b>K3</b>                          |
| <b>CO2</b>     | Integrate advanced techniques in approximation algorithms to tackle complex computational problems.                                                                     | <b>K3</b>                          |
| <b>CO3</b>     | Analyze the hardness of approximation for various problems.                                                                                                             | <b>K3</b>                          |
| <b>CO4</b>     | Design Fixed-parameter tractable (FPT) algorithms for practically relevant NP-hard problems.                                                                            | <b>K2</b>                          |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | 3   | 2   | -   | -   | -   | -   | -   | -    | 2    | 2    |
| <b>CO2</b> | 3   | 3   | 3   | 2   | 2   | -   | -   | -   | -   | -    | 2    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 3   | 2   | 2   | -   | -   | -   | -    | 2    | 2    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 3   | 2   | 2   | -   | -   | -    | 2    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                        |                                         |                              |                         |
|-------------------|----------------------------------------|-----------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>               | <b>Name of the Author/s</b>             | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | The Design of Approximation Algorithms | David P. Williamson,<br>David B. Shmoys | Cambridge University Press   | 1st Edition, 2011       |
| 2                 | Approximation Algorithms               | Vijay V. Vazirani                       | Springer                     | 1st Edition, 2001       |
| 3                 | Parameterized Algorithms               | Cygan et. al                            | Springer                     | 1st Edition, 2015       |

| <b>Reference Books</b> |                                                                                                        |                                                              |                              |                         |
|------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                                                               | <b>Name of the Author/s</b>                                  | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Randomized Algorithms                                                                                  | Rajeev Motwani,<br>Prabhakar Raghavan                        | Cambridge University Press   | 1st Edition, 1995       |
| 2                      | Approximation Algorithms for NP-Hard Problems                                                          | Dorit Hochbaum                                               | PWS Publishing               | 1st Edition, 1997       |
| 3                      | Complexity and Approximation: Combinatorial Optimization Problems and Their Approximability Properties | Giorgio Ausiello,<br>Pierluigi Crescenzi,<br>Giorgio Gambosi | Springer                     | 1st Edition, 2003       |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                          |
| 1                                     | <a href="https://onlinecourses.nptel.ac.in/noc24_cs97">https://onlinecourses.nptel.ac.in/noc24_cs97</a> |
| 2                                     | <a href="https://onlinecourses.nptel.ac.in/noc20_cs39">https://onlinecourses.nptel.ac.in/noc20_cs39</a> |
| 3                                     | <a href="https://onlinecourses.nptel.ac.in/noc21_cs92">https://onlinecourses.nptel.ac.in/noc21_cs92</a> |

## SEMESTER S7

### SOFTWARE QUALITY ASSURANCE

|                                            |                                          |                    |                |
|--------------------------------------------|------------------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT752</b>                          | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                                  | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                                        | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PEITT633: SOFTWARE<br>PROJECT MANAGEMENT | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To provide students with a deep understanding of Software Quality Assurance (SQA) concepts, and models, including McCall's Classic Model and ISO/IEC 25010, as well as the ability to evaluate software compliance with quality factors and navigate the challenges associated with maintaining software quality.
2. To enable students to design and implement effective SQA processes, including pre-project quality components, SQA plans, and project plans, while understanding the cost implications of software quality
3. To prepare students to apply various tools and methods, including templates, checklists, configuration management, and case tools/IDEs, to support SQA processes.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                        | <b>Contact Hours</b> |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Introduction:</b><br>SQA: Definitions and Concepts, McCall's Classic Model, ISO/IEC 25010 Model and other alternative models, Software compliance with quality factors, software quality challenges, Organization for assuring software quality, Components of SQA System.<br><b>Pre-Project Software Quality Components:</b><br>Contract Review, Development and Quality Plans | <b>9</b>             |

|          |                                                                                                                                                                                                                                                          |          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>2</b> | <b>SQA Process Implementation Plan and Activities:</b><br>Establishing SQA Process, SQA Plan and Project Plan, Cost of Software Quality, The SQA Model, SQA records and documentation control                                                            | <b>9</b> |
| <b>3</b> | <b>Product Assurance Activities for conformance:</b><br>Evaluation of Products, Reviews, Software Testing, Product Quality Metrics.<br><b>Process Assurance Activities:</b><br>Improvement Processes, Process Quality Metrics, Change Control Process.   | <b>9</b> |
| <b>4</b> | <b>Tools and Methods:</b><br>Templates and Checklists, Configuration Management, Case Tools and IDEs.<br><b>Process Standards:</b><br>Quality Assurance Standards, Quality Management Standards and Models, Project Progress Control, From SDLC to Agile | <b>9</b> |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Micro project</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|--------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                            | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

### End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| Part A                                                                                                                                                              | Part B                                                                                                                                                                                                                                                                            | Total     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 subdivisions.</li></ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                    | Bloom's Knowledge Level (KL) |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Explain the foundational concepts of Software Quality Assurance (SQA) and apply pre-project quality components to ensure software compliance with quality factors. | <b>K2</b>                    |
| <b>CO2</b>     | Develop and implement a comprehensive Software Quality Assurance (SQA) process                                                                                     | <b>K3</b>                    |
| <b>CO3</b>     | Describe product and process assurance activities to ensure software conformance.                                                                                  | <b>K2</b>                    |
| <b>CO4</b>     | Use tools and methods to effectively manage and improve software quality and familiarize with Agile methodologies.                                                 | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4      | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|----------|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   |          | 2   |     |     |     |     |      | 3    | 3    |
| <b>CO2</b> | 3   | 3   | 3   | 3        | 2   |     |     |     |     |      | 3    | 3    |
| <b>CO3</b> | 3   | 3   | 3   | <b>3</b> |     |     |     |     |     |      | 3    | 3    |
| <b>CO4</b> | 3   | 3   | 3   | 3        | 3   |     |     |     | 3   |      | 3    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                         |                             |                              |                                  |
|-------------------|-----------------------------------------|-----------------------------|------------------------------|----------------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b>          |
| 1                 | Software Quality: Concepts and Practice | Daniel Galin                | Wiley                        | 1 <sup>st</sup> edition,<br>2018 |

| <b>Reference Books</b> |                                                             |                             |                              |                                  |
|------------------------|-------------------------------------------------------------|-----------------------------|------------------------------|----------------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                    | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b>          |
| 1                      | Software Quality Assurance<br>From theory to implementation | Daniel Galin                | Pearson Education            | 2 <sup>nd</sup> edition,<br>2009 |
| 2                      | Software Engineering: A<br>Practitioner's Approach          | Roger S Pressman            | McGrawHill                   | 8 <sup>th</sup> edition,<br>2014 |
| 3                      | Software Quality: Theory and<br>Management                  | Alan Gillies                |                              | 3 <sup>rd</sup> edition,<br>2018 |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                              |
| <b>1</b>                              | <a href="https://nptel.ac.in/courses/106101061">https://nptel.ac.in/courses/106101061</a>                                   |
| <b>2</b>                              | <a href="https://onlinecourses.swayam2.ac.in/cec21_cs21/preview">https://onlinecourses.swayam2.ac.in/cec21_cs21/preview</a> |



## SEMESTER S7

### AUGMENTED AND VIRTUAL REALITY

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT753</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce the foundational concepts and technologies behind augmented and virtual reality systems.
2. To explore the development of AR and VR applications, focusing on interaction techniques and user experience.
3. To understand the hardware and software components used in AR and VR systems.
4. To examine the practical applications, challenges, and future trends in augmented and virtual reality.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                      | <b>Contact Hours</b> |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Introduction to Augmented and Virtual Reality-</b> Overview of AR and VR: Definitions, History, and Evolution, Basic Concepts in AR and VR: Immersion, Presence, Interaction, Components of AR and VR Systems: Display Devices, Input Devices, Tracking Systems, AR vs. VR: Differences and Similarities, Applications of AR and VR: Gaming, Education, Healthcare, Industry. | <b>9</b>             |
| <b>2</b>          | <b>AR and VR Application Development</b> - AR Application Development: SDKs and Tools (ARCore, ARKit) - VR Application Development: Game Engines (Unity, Unreal Engine) - Interaction Techniques: Gesture Recognition, Voice Commands, Haptic Feedback-User Experience (UX) Design in AR and VR: Design Principles, Best Practices-Case Studies:                                 | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | Successful AR and VR Applications.                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>3</b> | <b>AR and VR Hardware and Software</b> -AR Hardware: Smart Glasses, Head-Up Displays, Mobile Devices-VR Hardware: Head-Mounted Displays (HMDs), Motion Controllers, Treadmills-Software Architectures for AR and VR Systems-Rendering Techniques in AR and VR: Real-Time Graphics, 3D Audio, Physics Engines-Performance Optimization in AR and VR: Latency Reduction, Frame Rate Optimization. | <b>9</b> |
| <b>4</b> | <b>Advanced Topics and Applications in AR and VR</b> -Challenges in AR and VR: Technical, Ethical, and Social Issues-AR and VR in Education, Training, and Simulation-AR and VR in Healthcare: Therapy, Surgery, Rehabilitation-Future Trends in AR and VR: Mixed Reality (MR), Augmented Virtuality (AV), Metaverse-Ethical Considerations: Privacy, Security, and Impact on Society.          | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| <b>Part A</b>                                                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                                                                      | <b>Total</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

## Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                        | Bloom's Knowledge Level (KL) |
|----------------|------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Identify and describe the basic principles of augmented and virtual reality, including their systems and applications. | <b>K2</b>                    |
| <b>CO2</b>     | Demonstrate the ability to develop simple AR and VR applications with a focus on user interaction and experience       | <b>K2</b>                    |
| <b>CO3</b>     | Understand the appropriate hardware and software tools for building AR and VR systems.                                 | <b>K2</b>                    |
| <b>CO4</b>     | Discuss potential applications and future trends in AR and VR technology.                                              | <b>K2</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

## CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 2   | 2   | -   | -   | 2   | -   | -   | -   | -   | -    | -    | -    |
| CO2 | 2   | -   | 3   | 3   | 2   | -   | 2   | -   | -   | -    | -    | -    |
| CO3 | 2   | 2   | 3   | 3   | 2   | -   | -   | -   | -   | -    | -    | -    |
| CO4 | 2   | -   | -   | -   | 3   | 2   | 2   | 2   | -   | -    | -    | -    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                                                                                          |                                     |                             |                                 |
|------------|----------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------|---------------------------------|
| Sl. No     | Title of the Book                                                                                        | Name of the Author/s                | Name of the Publisher       | Edition and Year                |
| 1          | Augmented Reality: Principles and Practice                                                               | Dieter Schmalstieg, Tobias Hollerer | Addison-Wesley Professional | 1 <sup>st</sup> Edition<br>2016 |
| 2          | Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile | Tony Parisi                         | O'Reilly Media              | 1 <sup>st</sup> Edition<br>2015 |

| Reference Books |                                                                                                         |                                                       |                                |                                 |
|-----------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------|---------------------------------|
| Sl. No          | Title of the Book                                                                                       | Name of the Author/s                                  | Name of the Publisher          | Edition and Year                |
| 1               | Augmented Reality and Virtual Reality: The Power of AR and VR for Business                              | Timothy Jung, M.<br>Claudia tom Dieck                 | Springer                       | 1 <sup>st</sup> Edition<br>2021 |
| 2               | Virtual Reality and Augmented Reality: Myths and Realities                                              | Bruno Arnaldi, Pascal<br>Guitton, Guillaume<br>Moreau | CRC press                      | 1 <sup>st</sup> Edition<br>2018 |
| 3               | Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR | Steve Aukstakalnis                                    | Addison-Wesley<br>Professional | 1st Edition,<br>2017            |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                                     |
| 1                              | <a href="https://onlinecourses.swayam2.ac.in/nou23_ge34/preview">https://onlinecourses.swayam2.ac.in/nou23_ge34/preview</a> |
| 2                              | <a href="https://www.youtube.com/watch?v=zLMgdYI82IE">https://www.youtube.com/watch?v=zLMgdYI82IE</a>                       |
| 3                              | <a href="https://www.youtube.com/watch?v=MGuSTAqlZ9Q">https://www.youtube.com/watch?v=MGuSTAqlZ9Q</a>                       |

## SEMESTER S7

### NETWORK SCIENCE

|                                            |                                                                                            |                    |                |
|--------------------------------------------|--------------------------------------------------------------------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT754</b>                                                                            | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                                                                                    | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                                                                                          | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | GAMAT301-Mathematics for Information Science, GAMAT401-Mathematics for Information Science | <b>Course Type</b> | Theory         |

#### Course Objectives:

4. The students will learn the main concepts and characteristics of network science.
5. The students will learn the evolution of random networks based on graph theoretical concepts.
6. To equip the students to analyse the robustness of the network by assessing the structural vulnerabilities.
7. To equip the students to model disease spreading patterns in communities.

#### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Contact Hours</b> |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Introduction to network science - The main premise of network science, History and relation to graph theory, physics, sociology, and other disciplines, Examples of networks from different application domains, The characteristics of Network Science, societal and scientific impact.<br><br>Familiarisation of Network analysis and visualization tools - NetworkX, Gephi, Cytoscape, Infomap, Igraph, Statnet, Network Workbench, Pajek network visualization, Jung network analysis, GraphViz, Matlab's Random Boolean Networks (RBN) toolbox. | <b>9</b>             |
| <b>2</b>          | Relevant Concepts From Graph Theory - Undirected, directed, signed, weighted and spatial networks, Paths, connected components, Directed Acyclic Graphs, Bipartite graphs, Clustering coefficients.<br><br>The Random Network Model - Introduction, Number of Links, Degree                                                                                                                                                                                                                                                                          | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | <p>Distribution in random network, The Evolution of a Random Network, Small Worlds, Clustering Coefficient in random network, Watts-Strogatz model.</p> <p>The Barabasi-Albert model- Growth and Preferential Attachment, The Barabási-Albert Model.</p>                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>3</b> | <p>Network Robustness- Percolation Theory, Inverse Percolation Transition and Robustness, Robustness of Scale-free Networks, Molloy-Reed Criterion, Critical Threshold, Attack Tolerance, Cascading Failures, Modelling Cascading Failures, Failure Propagation Model, Branching Model, Building Robustness, Designing Robust Networks.</p>                                                                                                                                                                                                                                                                                    | <b>9</b> |
| <b>4</b> | <p>Communities- Basics of Communities, Hierarchical Clustering, Agglomerative Procedures: the Ravasz Algorithm, Divisive Procedures: the Girvan-Newman Algorithm, Hierarchy in Real Networks, Modularity.</p> <p>Spreading Phenomena- Epidemic Modelling, Susceptible-Infected (SI) Model, Susceptible-Infected-Susceptible (SIS) Model, Susceptible-Infected-Recovered (SIR) Model, Network Epidemics, Susceptible-Infected (SI) Model on a Network, SIS Model and the Vanishing Epidemic Threshold, Contact Networks, Digital Viruses, Immunization, Random Immunization, Vaccination Strategies in Scale-Free Networks.</p> | <b>9</b> |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

### End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| Part A                                                                                                                                                      | Part B                                                                                                                                                                                                                                                                      | Total     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p>(8x3 =24marks)</p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 sub divisions.</li></ul> <p>(4x9 = 36 marks)</p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                              | Bloom's Knowledge Level (KL) |
|----------------|------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Discuss the main concepts and characteristics of network science             | <b>K2</b>                    |
| <b>CO2</b>     | Explain the evolution of random network based on graph theoretical concepts. | <b>K2</b>                    |
| <b>CO3</b>     | Build robust network by assessing the structural vulnerabilities             | <b>K3</b>                    |
| <b>CO4</b>     | Model disease spreading pattern in communities.                              | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   |     | -   | -   |     | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO2</b> | 2   | 3   | 2   | -   | -   | -   | -   | -   | -   | -    | -    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 3   | 3   | 2   | -   | -   | -   | -    | -    | 2    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 3   | 3   | -   | -   | -   | -    | -    | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                             |                      |                            |                  |
|------------|-----------------------------|----------------------|----------------------------|------------------|
| Sl. No     | Title of the Book           | Name of the Author/s | Name of the Publisher      | Edition and Year |
| 1          | Network Science             | A-L. Barabási        | Cambridge University Press | 2016             |
| 2          | Networks - An introduction_ | M.E.J. Newman        | Oxford Univ Press          | 2010             |

| Reference Books |                                                       |                                         |                            |                  |
|-----------------|-------------------------------------------------------|-----------------------------------------|----------------------------|------------------|
| Sl. No          | Title of the Book                                     | Name of the Author/s                    | Name of the Publisher      | Edition and Year |
| 1               | Complex Networks - Structure, Robustness and Function | R. Cohen and S. Havlin                  | Cambridge University Press | 2010             |
| 2               | Social and Economic Networks                          | M.O. Jackson                            | Princeton Univ Press       | 2008             |
| 3               | Dynamical Processes on Complex Networks               | Barrat, M. Barthelemy and A. Vespignani | Cambridge Univ Press       | 2008             |
| 4               | Statistical analysis of network data                  | E. Kolaczyk                             | Springer                   | 2009             |

| Video Links (NPTEL, SWAYAM...) |                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                   |
| 1                              | <a href="https://nptel.ac.in/courses/106105154">https://nptel.ac.in/courses/106105154,</a>                |
| 2                              | <a href="https://www.youtube.com/watch?v=oqOyTdLsq3o">https://www.youtube.com/watch?v=oqOyTdLsq3o</a>     |
| 3                              | <a href="https://www.youtube.com/watch?v=2pnLsvz1fSw">https://www.youtube.com/watch?v=2pnLsvz1fSw,</a>    |
| 4                              | <a href="https://www.youtube.com/watch?v=WABtTfTnVCI">https://www.youtube.com/watch?v=WABtTfTnVCI</a>     |
| 5                              | <a href="https://www.youtube.com/watch?v=Q7M79w1ZAY8">https://www.youtube.com/watch?v=Q7M79w1ZAY8</a>     |
| 6                              | <a href="https://www.youtube.com/watch?v=DmwFt5QjL1s">https://www.youtube.com/watch?v=DmwFt5QjL1s</a>     |
| 7                              | <a href="https://www.youtube.com/watch?v=TYod3lGidPo">https://www.youtube.com/watch?v=TYod3lGidPo</a>     |
| 8                              | <a href="https://www.youtube.com/watch?v=W8sR_sq_Lec">https://www.youtube.com/watch?v=W8sR_sq_Lec, ,,</a> |
| 9                              | <a href="https://www.youtube.com/watch?v=qEjhXvqADTQ">https://www.youtube.com/watch?v=qEjhXvqADTQ,</a>    |
| 10                             | <a href="https://www.youtube.com/watch?v=X3xE9oj5-3o">https://www.youtube.com/watch?v=X3xE9oj5-3o</a>     |



**SEMESTER S7**  
**CYBER AND NETWORK FORENSICS**

|                                            |                             |                    |                |
|--------------------------------------------|-----------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEITT755</b>             | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                     | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 5/3                         | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PCITT402: Computer Networks | <b>Course Type</b> | Theory         |

**Course Objectives:**

1. To introduce the fundamental concepts of cyber forensics and its importance in investigating cybercrimes.
2. To explore the techniques and tools used in network forensics for detecting and analysing network intrusions.
3. To understand the legal aspects, standards, and procedures involved in cyber and network forensics investigations.
4. To examine advanced topics in cyber and network forensics, including incident response and digital evidence management.

**SYLLABUS**

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                          | <b>Contact Hours</b> |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Introduction to Cyber Forensics:<br>Overview of Cyber Forensics: Definition, Importance, and Applications<br>Types of Cybercrimes: Hacking, Phishing, Identity Theft, Cyberstalking<br>Digital Evidence: Collection, Preservation, and Analysis<br>Cyber Forensics Tools: EnCase, FTK, Autopsy<br>Case Studies: Real-World Cybercrime Investigations | <b>9</b>             |
| <b>2</b>          | Network Forensics:<br>Introduction to Network Forensics: Definition and Importance<br>Network Traffic Analysis: Packet Sniffing, Protocol Analysis, Flow Analysis                                                                                                                                                                                    | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                     |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | Intrusion Detection and Prevention Systems: Snort, Suricata<br>Network Forensics Tools: Wireshark, tcpdump, NetFlow<br>Case Studies: Network Breach Investigations                                                                                                                                                                                                                                  |          |
| <b>3</b> | Legal Framework for Cybercrime Investigations: Laws and Regulations<br>Digital Forensics Standards and Best Practices: ISO/IEC 27037<br>Chain of Custody: Importance, Documentation, and Maintenance<br>Report Writing and Presentation in Forensics: Documentation, Expert Testimony<br>Ethical Issues in Cyber and Network Forensics                                                              | <b>9</b> |
| <b>4</b> | Advanced Topics in Cyber and Network Forensics:<br>Incident Response: Preparation, Detection, Containment, Eradication, Recovery<br>Digital Evidence Management: Storage, Archiving, and Retrieval<br>Cloud Forensics: Challenges and Solutions<br>Forensics in Mobile Devices and IoT: Techniques and Tools<br>Future Trends in Cyber and Network Forensics: AI in Forensics, Blockchain Forensics | <b>9</b> |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Internal Examination (Written)</b> | <b>Evaluate Level Assessment</b> | <b>Analyze Level Assessment</b> | <b>Total</b> |
|-------------------|---------------------------------------|----------------------------------|---------------------------------|--------------|
| <b>5</b>          | <b>15</b>                             | <b>10</b>                        | <b>10</b>                       | <b>40</b>    |

**Evaluate and Analyze Level Assessment [20 Marks]**

Students should evaluate and analyze a real-world problem, assess the proposed solutions, provide a conclusion on which solution is most appropriate for the problem, and implement the chosen solution.

**Criteria for evaluation:**

- **Problem Definition (K4 - 4 points)**
  - *Clearly defines the real-world problem.*
  - *Collects and preserves evidence meticulously, following all legal and procedural guidelines.*
- **Problem Analysis (K4 - 4 points)**
  - *Break-down and presents a well-reasoned solution approach.*
  - *Provides insightful and accurate analysis of the evidence, drawing well-supported conclusions..*
- **Evaluate (K5 - 4 points)**
  - *Demonstrates exceptional understanding and application of advanced forensic tools and techniques.*
  - *Shows a strong commitment to ethical standards, ensuring privacy and legal compliance throughout the project.*
- **Implementation (K5 - 4 points)**
  - *Demonstrates expert handling of incidents with clear, effective, and timely response strategies.*
  - *Demonstrates exceptional creativity and problem-solving skills*
- **Conclusion (K4- 2 points, K5 – 2 points)**
  - *Summarizes findings and insights. (K4)*
  - *Reflects critical thinking and informed decision-making. (K5)*

**Scoring:**

- **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
- **Competent (3 points):** Solid performance with minor areas for improvement.
- **Developing (2 points):** Adequate effort but lacks depth or clarity.
- **Minimal (1 point):** Incomplete or significantly flawed.

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| <b>Part A</b>                                                                                                                                                      | <b>Part B</b>                                                                                                                                                                                                                                                                      | <b>Total</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 sub divisions.</li></ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| <b>Course Outcome</b> |                                                                                                                         | <b>Bloom's Knowledge Level (KL)</b> |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>CO1</b>            | Apply the basic principles of cyber forensics to investigate and analyze digital evidence.                              | <b>K3</b>                           |
| <b>CO2</b>            | Implement network forensics techniques to detect and analyze network intrusions.                                        | <b>K3</b>                           |
| <b>CO3</b>            | Utilize legal standards and procedures in conducting cyber and network forensics investigations                         | <b>K3</b>                           |
| <b>CO4</b>            | Analyse and manage digital evidence in advanced cyber forensics scenarios.                                              | <b>K4</b>                           |
| <b>CO5</b>            | Synthesize and evaluate comprehensive cyber and network forensics strategies for large-scale and complex investigations | <b>K5</b>                           |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 2   | 3   | 2   | -   | -   | 3   | 2   | 2    | 2    | 3    |
| <b>CO2</b> | 3   | 3   | 3   | 3   | 2   | 2   | -   | 2   | 2   | 2    | 2    | 3    |
| <b>CO3</b> | 3   | 3   | 2   | 2   | -   | 2   | 3   | 3   | 2   | 3    | 3    | 2    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 2   | 2   | -   | 2   | 2   | 2    | 3    | 3    |
| <b>CO5</b> | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3    | 3    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                        |                                   |                              |                         |
|-------------------|--------------------------------------------------------|-----------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                               | <b>Name of the Author/s</b>       | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Digital Forensics and Cyber Crime: A Hands-On Approach | Nilakshi Jain, Dhananjay Kalbande | Wiley                        | 11st Edition, 2017      |

| <b>Reference Books</b> |                                                                     |                                        |                              |                         |
|------------------------|---------------------------------------------------------------------|----------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                            | <b>Name of the Author/s</b>            | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Computer Forensics: Investigating Network Intrusions and Cybercrime | EC-Council                             | Cengage Learning             | 2nd Edition, 2010       |
| 2                      | Network Forensics: Tracking Hackers through Cyberspace              | Sherri Davidoff, Jonathan Ham          | Prentice Hall                | 1st Edition, 2012       |
| 3                      | Legal Principles of Digital Forensics                               | Kim-Kwang Raymond Choo                 | Springer                     | 1st Edition, 2019       |
| 4                      | Incident Response & Computer Forensics                              | Kevin Mandia, Chris Prosise, Matt Pepe | McGraw-Hill Education        | 3rd Edition, 2014       |

| Video Links (NPTEL, SWAYAM...) |                                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                               |
| 1                              | <a href="https://www.youtube.com/watch?v=PgS8XHnOzbU">https://www.youtube.com/watch?v=PgS8XHnOzbU</a> |
| 2                              | <a href="https://www.youtube.com/watch?v=8wF5WysAUzU">https://www.youtube.com/watch?v=8wF5WysAUzU</a> |
| 3                              | <a href="https://www.youtube.com/watch?v=3FZMy5TJ4HQ">https://www.youtube.com/watch?v=3FZMy5TJ4HQ</a> |
| 4                              | <a href="https://www.youtube.com/watch?v=93-hn8QIgr8">https://www.youtube.com/watch?v=93-hn8QIgr8</a> |

## SEMESTER S7

### MACHINE LEARNING

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEITT721</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To provide students the overview of machine learning, including various types of learning, data preprocessing techniques, and key algorithms.
2. To enable students to acquire the knowledge of model selection, regularization techniques and neural network architectures.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                   | <b>Contact Hours</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Introduction to Machine Learning: Overview of Machine Learning- Definition, Types of Machine Learning- Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data Preprocessing- Cleaning, Normalization and Splitting Data-Training St, Test Set and Validation Set, Feature Engineering- Feature Selection and Extraction. | <b>9</b>             |
| <b>2</b>          | Supervised Learning Algorithms: Linear Models- Linear Regression and Logistic Regression, Decision Trees- Basics, Splitting Criteria and Overfitting, Ensemble Methods- Random Forests and Gradient Boosting Machines, Support Vector Machines (SVMs)- SVM Concepts and Kernels.                                                              | <b>11</b>            |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>3</b> | Model Selection and Neural Networks: Model Evaluation Metrics- Accuracy, Precision, Recall, F1 Score, Receiver Operating Characteristic Curve (ROC Curve), Area Under ROC Curve (AUC Curve), Model Selection- Cross-Validation and Bias-Variance Trade-off, Regularization Techniques- L1 and L2 Regularization, Introduction to Neural Networks- Perceptron, Multi-Layer Perceptron and Perceptron Training Algorithm. | <b>9</b> |
| <b>4</b> | Unsupervised Learning Algorithms: Clustering- K-Means, Hierarchical Clustering and Density-based Spatial Clustering of Applications with Noise (DBSCAN), Dimensionality Reduction- Principal Component Analysis (PCA), Reinforcement Learning (RL)- RL Concepts and Q-Learning.                                                                                                                                         | <b>7</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| <b>Part A</b>                                                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                                                                      | <b>Total</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |



### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                                               | Bloom's Knowledge Level (KL) |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Explain machine learning concepts and gain insights into data preprocessing techniques and feature engineering.                                                                               | <b>K2</b>                    |
| <b>CO2</b>     | Demonstrate proficiency in implementing supervised learning algorithms and understand the key concepts in SVMs.                                                                               | <b>K3</b>                    |
| <b>CO3</b>     | Gain insights into model selection and architecture of neural networks as well as apply cross-validation techniques to address bias-variance trade-offs and implement regularization methods. | <b>K3</b>                    |
| <b>CO4</b>     | Explore and apply unsupervised learning techniques and reinforcement learning.                                                                                                                | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | 2   | 2   | -   | -   | 2   | -   | -   | -    | -    | 2    |
| <b>CO2</b> | 3   | 3   | 2   | 2   | -   | -   | 2   | -   | -   | -    | -    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 2   | -   | -   | 2   | -   | -   | -    | -    | 2    |
| <b>CO4</b> | 3   | 2   | 2   | 2   | -   | -   | 2   | -   | -   | -    | -    | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                               |                             |                              |                         |
|-------------------|-----------------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                      | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Pattern Recognition and Machine Learning      | Bishop, C. M.               | Springer, New York           | 2006                    |
| 2                 | Introduction to Machine Learning              | E. Alpaydin                 | PHI                          | 2005                    |
| 3                 | Machine Learning                              | Tom Mitchell                | McGraw Hill                  | 1997                    |
| 4                 | Machine Learning: A Probabilistic Perspective | Kevin P. Murphy             | MIT Press                    | 2012                    |

| <b>Reference Books</b> |                                      |                                             |                              |                         |
|------------------------|--------------------------------------|---------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>             | <b>Name of the Author/s</b>                 | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | The Elements of Statistical Learning | Hastie, T., Tibshirani, R. and Friedman, J. | Springer                     | 2001                    |
| 2                      | Introduction to Machine Learning     | Alex Smola and SVN                          | Cambridge University Press   | 2008                    |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                          |
| 1                                     | <a href="https://onlinecourses.nptel.ac.in/noc22_cs97/preview">https://onlinecourses.nptel.ac.in/noc22_cs97/preview</a> |
| 2                                     | <a href="https://onlinecourses.nptel.ac.in/noc24_ma87/preview">https://onlinecourses.nptel.ac.in/noc24_ma87/preview</a> |

## SEMESTER S7

### DATA SCIENCE FOR ENGINEERS

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEITT722</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. **Develop proficiency in R programming** by mastering foundational concepts such as variables, data types, and data manipulation techniques, including data frames, recasting, and joining. Students will also gain experience in implementing functions, control structures, and performing arithmetic, logical, and matrix operations in R, enabling them to handle complex data analysis tasks.
2. **Apply key data science techniques** across linear algebra, statistics, supervised, and unsupervised learning. This includes solving linear equations, understanding statistical modeling, executing linear and logistic regression, and performing clustering methods like k-means and hierarchical clustering, all with practical implementation in R.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                        | <b>Contact Hours</b> |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>R for Data Science</b> - Introduction to R, Variables, and Datatypes in R, Data Frames, Recasting and Joining of Dataframes, Arithmetic, Logical and Matrix operations in R, Functions, Control structures, Data visualization in R.                                                            | <b>9</b>             |
| <b>2</b>          | <b>Linear Algebra for Data Science</b> - Solving Linear Equations, Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors.<br><b>Statistics for Data Science</b> - Statistical Modelling, Random Variables and Probability Mass/Density Functions, Sample Statistics, Hypotheses Testing, | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>3</b> | <b>Foundations of Data Science</b> - Introduction to data science, Properties of data, Classification of Data Science, Data Science Process.<br><b>Supervised Learning</b> - Regression - Simple Linear Regression, Multiple Linear Regression, Model Assessment, Implementation of Linear Regression Models in R, Cross Validation, Classification - Logistic Regression, Performance Measures, Logistic Regression Implementation in R, k - Nearest Neighbors (kNN), k - Nearest Neighbors Implementation in R. | <b>11</b> |
| <b>4</b> | <b>Unsupervised Learning</b> – Clustering - k- means Clustering, k-means implementation in R, Hierarchical Clustering – Agglomerative Clustering Algorithm, Implementation in R.                                                                                                                                                                                                                                                                                                                                  | <b>7</b>  |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE):**

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

| <b>Part A</b>                                                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                          | <b>Total</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <p>4. Each question carries 9 marks.</p> <p>5. Two questions will be given from each module, out of which 1 question should be answered.</p> <p>6. Each question can have a maximum of 3 sub divisions.</p> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

## Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                                                                                                                            | Bloom's Knowledge Level (KL) |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Utilize R for data science, including variable management, data frames manipulation, recasting, and joining operations, as well as performing arithmetic, logical, and matrix operations using R.                                                                          | <b>K2</b>                    |
| <b>CO2</b>     | Discuss linear algebra concepts crucial to data science, such as solving linear equations, understanding distances, hyperplanes, halfspaces, and calculating eigenvalues and eigenvectors.                                                                                 | <b>K2</b>                    |
| <b>CO3</b>     | Explain statistical modeling, random variables, probability distributions, and hypotheses testing, and will be able to apply these concepts to real-world data science problems.                                                                                           | <b>K2</b>                    |
| <b>CO4</b>     | Implement various supervised learning models, such as linear regression and logistic regression, and unsupervised learning techniques, such as k-means and hierarchical clustering, using R, and will be able to assess model performance and validate results effectively | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

## CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | 2   | 2   | 3   | -   | -   | -   | 2   | -    | -    | 2    |
| <b>CO2</b> | 3   | 3   | 2   | 2   | 2   | -   | -   | -   | 2   | -    | -    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 2   | 2   | -   | -   | -   | 2   | -    | -    | 3    |
| <b>CO4</b> | 3   | 2   | 2   | 2   | 3   | -   | -   | -   | 2   | -    | -    | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                                     |                                               |                       |                                     |
|------------|-----------------------------------------------------|-----------------------------------------------|-----------------------|-------------------------------------|
| Sl. No     | Title of the Book                                   | Name of the Author/s                          | Name of the Publisher | Edition and Year                    |
| 1          | Data Science for Engineers                          | Raghunathan<br>Rengaswamy, Resmi<br>Suresh    | CRC Press             | 1 <sup>st</sup><br>Edition,<br>2022 |
| 2          | Applied Statistics and<br>Probability for Engineers | Douglas C. Montgomery<br>and George C. Runger | Wiley                 | 6 <sup>th</sup><br>Edition,<br>2016 |
| 3          | Data science: Concepts and<br>practice              | Kotu, V., & Deshpande, B                      | Morgan Kaufmann       | 2019                                |

| Reference Books |                                                                                                       |                                                      |                           |                                  |
|-----------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------|----------------------------------|
| Sl. No          | Title of the Book                                                                                     | Name of the Author/s                                 | Name of the Publisher     | Edition and Year                 |
| 1               | Data Science and Big Data<br>Analytics: Discovering,<br>Analyzing, Visualizing and<br>Presenting Data | David Dietrich,<br>Barry Heller and<br>Beibei Yang   | EMC Education<br>Services | 1 <sup>st</sup> Edition,<br>2015 |
| 2               | An Introduction to Statistical<br>Learning: with Applications in<br>R.                                | James, G., Witten, D.,<br>Hastie, T., Tibshirani, R. | Springer                  | 2017                             |
| 3               | R for Data Science: Import,<br>Tidy, Transform, Visualize, and<br>Model Data                          | Hadley Wickham,<br>Garrett Grolemund                 | O'Reilly                  | 2017                             |

| Video Links (NPTEL, SWAYAM...) |                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                   |
| 1                              | <a href="https://nptel.ac.in/courses/106106179">https://nptel.ac.in/courses/106106179</a> |

## SEMESTER S7

### INTERNET OF THINGS

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEITT723</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce the fundamental concepts, architecture, and protocols of the Internet of Things (IoT).
2. To explore the hardware and software platforms used in IoT systems.
3. To understand the role of data analytics and cloud integration in IoT.
4. To examine the applications, security challenges, and future trends in IoT.

#### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>Contact Hours (36 Hrs)</b> |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>1</b>          | Introduction to IoT:<br>Internet, Intranet and Extranet. Comparison,<br>Internet of Things Vision, IoT Strategic Research and Innovation Directions,<br>IoT Applications, Infrastructure, Networks and Communication, Processes,<br>Data Management, Security, Privacy & Trust, Device Level Energy Issues.<br>IoT reference model, IoT Architecture and core IoT function, Layers,<br>Components, and Protocols. | <b>8</b>                      |
| <b>2</b>          | <b>IoT Protocols:</b><br><br>Physical Data Link Layer: IEEE 802.15.4, LoRa, NFC, Zigbee,<br>Bluetooth/Bluetooth Low Energy (BLE), 6LoWPAN<br><br>Network Layer: IPv4/IPv6, RPL, CoAP                                                                                                                                                                                                                              | <b>8</b>                      |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          | <p>Transport Layer: TCP, UDP, DTLS</p> <p>Session &amp; Application Layer: CoAP, MQTT, HTTP/HTTPS, XMPP, AMQP, DDS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |
| <b>3</b> | <p>IoT Hardware and Software Platforms :</p> <p>IoT Hardware Platforms: Sensors -LED, Switch, LDR, Temperature Sensor, Fiber Optic.</p> <p>Actuators and Smart Objectives. Raspberry Pi Development Kit, Arduino IDE and board types, ESP8266</p> <p>IoT Software Platforms: Node-RED, ThingSpeak, Blynk, IoTivity</p> <p>IoT Operating Systems: Contiki, RIOT, TinyOS</p> <p>Cloud Platforms for IoT: AWS IoT, Google Cloud IoT, Microsoft Azure.</p> <p>Introduction to Data Analytics in IoT: Importance, Techniques, and Tools</p> <p>Data Storage and Management in IoT: Edge Computing, Fog Computing</p> <p>Data Visualization and Reporting in IoT</p> | <b>11</b> |
| <b>4</b> | <p>Building IoT Applications: Data Collection, Processing, and Actuation, Security.</p> <p>IoT Security Challenges: Threats, Vulnerabilities, and Solutions.</p> <p>IoT Privacy and Data Protection: Best Practices and Standards</p> <p>Future Trends in IoT: AI Integration, 5G, IoT for Sustainability</p> <p>Case Studies: Smart Homes, Smart Parking , Transportation, Wearables. Industrial IoT-Smart Agriculture, Smart Cities, Healthcare IoT, Smart and connected Cities- Street layer, City Layer, Data Center Layer.</p>                                                                                                                            | <b>9</b>  |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |



**End Semester Examination Marks (ESE):**

| In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions                                        |                                                                                                                                                                                                                                                                                        |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Part A                                                                                                                                                                | Part B                                                                                                                                                                                                                                                                                 | Total     |
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                              | Bloom's Knowledge Level (KL) |
|----------------|----------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Apply the basic principles of IoT to understand its architecture, components, and protocols. | <b>K3</b>                    |
| <b>CO2</b>     | Develop IoT applications using appropriate hardware and software platforms.                  | <b>K3</b>                    |
| <b>CO3</b>     | Integrate data analytics and cloud services in IoT systems for enhanced functionality.       | <b>K3</b>                    |
| <b>CO4</b>     | Analyse the security challenges and future trends in IoT applications.                       | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 2   | 2   | 2   | -   | -   | 2   | -   | -    | 2    | 1    |
| <b>CO2</b> | 3   | 3   | 3   | 3   | 3   | -   | -   | 2   | -   | 2    | 2    | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 3   | 3   | 2   | -   | 2   | 1   | 2    | 2    | 3    |
| <b>CO4</b> | 3   | 3   | 2   | 3   | 2   | 3   | 2   | 3   | -   | -    | 3    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                         |                                 |                       |                   |
|------------|-----------------------------------------|---------------------------------|-----------------------|-------------------|
| Sl. No     | Title of the Book                       | Name of the Author/s            | Name of the Publisher | Edition and Year  |
| 1          | Internet of Things: A Hands-On Approach | Arshdeep Bahga, Vijay Madisetti | Universities Press    | 1st Edition, 2015 |

| Reference Books |                                                                                                                  |                                                  |                       |                              |
|-----------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------|------------------------------|
| Sl. No          | Title of the Book                                                                                                | Name of the Author/s                             | Name of the Publisher | Edition and Year             |
| 1               | The Internet of Things: Key Applications and Protocols                                                           | Olivier Hersent, David Boswarthick, Omar Elloumi | Wiley                 | 2nd Edition, 2012            |
| 2               | Building the Internet of Things: Implement New Business Models, Disrupt Competitors, and Transform Your Industry | Maciej Kranz                                     | Wiley                 | 1st Edition, 2016            |
| 3               | Cloud Computing and the Internet of Things: Technologies, Applications and Security                              | Amita Kapoor, Arshdeep Bahga, Vijay Madisetti    | Springer              | 1st Edition, 2020            |
| 4               | Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations                       | Fei Hu                                           | CRC Press             | 1st Edition, 2016            |
| 5               | Arduino Cookbook                                                                                                 | Michael Margolis                                 | O'Reilly Media        | 3 <sup>rd</sup> Edition 2020 |
| 6               | Getting Started With Raspberry Pi                                                                                | Matt Richardson, Shawn Wallace                   | O'Reilly Media        | 3 <sup>rd</sup> Edition 2013 |

| Video Links (NPTEL, SWAYAM...) |                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|
| No.                            | Link ID                                                                                                       |
| 1                              | <a href="https://onlinecourses.nptel.ac.in/noc24_cs115/">https://onlinecourses.nptel.ac.in/noc24_cs115/</a>   |
| 2                              | <a href="https://onlinecourses.swayam2.ac.in/ntr24_ed44/">https://onlinecourses.swayam2.ac.in/ntr24_ed44/</a> |
| 3                              | <a href="https://onlinecourses.nptel.ac.in/noc24_cs115">https://onlinecourses.nptel.ac.in/noc24_cs115</a>     |

# **SEMESTER 8**

**INFORMATION TECHNOLOGY**